Z-GRADED PRIMES OF A RING WITH A HOMOGENEOUS UNIT MAP HOMEOMORPHICALLY TO THE SPECTRUM OF THE DEGREE-ZERO PART

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Allowed standard prerequisites (editorial).

- Homogeneous invertibility and the operators dividing degrees by the unit. Used throughout the counted proof (lines 13637-13645). Lets a homogeneous relation be rescaled by powers of the invertible element so that its degree-zero part becomes visible.
- Basic open sets and their behaviour under the inverse map. Used in the second half of the counted proof (lines 13650-13653). Shows that images and preimages of basic opens are open, which upgrades the bijection to a homeomorphism.

Lemma 0.1. *Let S be a $\mathbf{Z}$-graded ring containing a homogeneous invertible element of positive degree. Then the set $G \subset \mathrm{Spec}(S)$ of $\mathbf{Z}$-graded primes of S (with induced topology) maps homeomorphically to $\mathrm{Spec}(S_0)$.*

Proof. First we show that the map is a bijection by constructing an inverse. Let $f \in S_d$, $d > 0$ be invertible in S. If $\mathfrak{p}_0$ is a prime of S_0, then $\mathfrak{p}_0 S$ is a $\mathbf{Z}$-graded ideal of S such that $\mathfrak{p}_0 S \cap S_0 = \mathfrak{p}_0$. And if $ab \in \mathfrak{p}_0 S$ with a, b homogeneous, then $a^d b^d / f^{\deg(a)+\deg(b)} \in \mathfrak{p}_0$. Thus either $a^d / f^{\deg(a)} \in \mathfrak{p}_0$ or $b^d / f^{\deg(b)} \in \mathfrak{p}_0$, in other words either $a^d \in \mathfrak{p}_0 S$ or $b^d \in \mathfrak{p}_0 S$. It follows that $\sqrt{\mathfrak{p}_0 S}$ is a $\mathbf{Z}$-graded prime ideal of S whose intersection with S_0 is $\mathfrak{p}_0$.

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To show that the map is a homeomorphism we show that the image of $G \cap D(g)$ is open. If $g = \sum g_i$ with $g_i \in S_i$, then by the above $G \cap D(g)$ maps onto the set $\bigcup D(g_i^d/f^i)$ which is open. $\square$